Paleomagnetism of Middle Proterozoic diabase sheets from central Arizona

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Paleomagnetic results from 1090 Ma diabase sheets of the southwestern U.S.A. diabase province exposed in central Arizona yield two distinct remanent magnetizations (herein termed ADn and ADr), in accordance with the findings of previous investigations. Magnetization ADn is well-defined and has an in situ mean direction of $D = 283.3°$, $I = 45.1°$ ($k = 17.7$, $\alpha_{95} = 8.7°$, $n = 17$ independent observations). A mean pole, after correction of paleomagnetic site means for a net 5° clockwise rotation of the Colorado Plateau and transition zone, is located at 22.7°N, 179.3°E ($K = 21.9$, $A_{95} = 7.8°$). The second magnetization (ADr) gives an in situ mean direction of $D = 161.1°$, $I = -87.5°$ ($k = 22.2$, $\alpha_{95} = 19.9°$, $n = 4$ independent observations) with a poorly defined pole at 37.6°N, 247.6°E ($K = 6.5$, $A_{95} = 38.9°$). Rock magnetic and alternating field and thermal demagnetization characteristics indicate the ADn and ADr magnetizations are both carried by low-Ti titanomagnetite. Both magnetizations are interpreted to be primary thermoremanent magnetizations acquired during emplacement and cooling of the diabase sheets at about 1090–1100 Ma. Comparison of the ADn pole and published geochronologic data from the Arizona diabase with the well-dated normal polarity poles of the Keweenawan region indicates that mafic magmatism in the southwestern U.S.A. diabase province and in the midcontinent rift was essentially synchronous.

Les résultats paléomagnétiques des couches de diabase de 1090 Ma de la région sud-ouest de la province de diabase U.S.A., exposées dans le centre de l'Arizona, donnent deux aimantations rémanentes distinctes (ici désignées par ADn et ADr), conformes aux résultats obtenus dans des études antérieures. L'aimantation ADn est clairement définie et elle possède une direction in situ $D = 283{,}3°$, $I = 45{,}1°$ ($k = 17{,}7$, $\alpha_{95} = 8{,}7°$, $n = 17$ observations indépendantes). Un pôle moyen, après correction appliquée aux moyennes des sites paléomagnétiques pour une rotation nette de 5° dans le sens des aiguilles d'une montre du Plateau du Colorado et de la zone de transition, est localisé à 22,7°N, 179,3°E ($K = 21{,}9$, $A_{95} = 7{,}8°$). La deuxième aimantation (ADr) donne une direction in situ $D = 161{,}1°$, $I = -87{,}5°$ ($k = 22{,}2$, $\alpha_{95} = 19{,}9°$, $n = 4$ observations indépendantes) avec un pôle vaguement localisé à 37,6°N, 247,6°E ($K = 6{,}5$, $A_{95} = 38{,}9°$). Les caractéristiques de la roche magnétique et de la désaimantation par champ alternatif et par chauffage indiquent que les deux aimantations ADn et ADr sont portées par la titanomagnétite pauvre en Ti. Les deux aimantations sont interprétées comme étant des aimantations thermorémanentes primaires acquises durant les événements de mise en place et de refroidissement des couches de diabase, il y a environ 1090 à 1100 Ma. La comparaison du pôle ADn et des données géochronologiques publiées pour la diabase d'Arizona avec les pôles de polarité normale datés adéquatement de la région de Keweenavan révèle que le magmatisme mafique dans le sud-ouest de la province de diabase U.S.A. et l'existence du rift midcontinental étaient essentiellement synchrones.

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Introduction

In this report, we present new paleomagnetic results from diabase sheets exposed in Salt River Canyon and adjacent areas of central Arizona (Fig. 1). Early paleomagnetic results from this region (Helsley 1965; Helsley and Spall 1972) demonstrated the presence of a normal polarity magnetization of northwest declination and moderate positive inclination whose paleomagnetic pole was consistent with those of similar age from the Keweenawan region of the midcontinent rift. Recently, however, Sandberg and Butler (1986) found a pre-Laramide deformation (i.e., greater than 50–70 Ma) magnetization of steep negative inclination from a sill in the bottom of Salt River Canyon. The purpose of this study was to reinvestigate the paleomagnetism of the Arizona diabase using modern methods and to identify, if possible, additional sheets of negative inclination in order to better understand the origin of this magnetization. Our results confirm the results of both studies and document additional sheets that carry a reversed magnetization of steep negative inclination. Combined paleomagnetic and geochronologic data from the Arizona diabase and Keweenawan rocks indicate that mafic magmatism in the southwestern U.S.A. diabase province and the midcontinent rift was synchronous within the limits of available data.

Geologic setting

Middle Proterozoic mafic intrusions of the southwestern U.S.A. diabase province are a ubiquitous feature of the Precambrian geology of central and western Arizona and southeastern California. Extensive, thick (up to 400 m), sometimes composite, sheet-like bodies intrude 1.4 Ga anorogenic granites and 1.6–1.7 Ga metamorphic rocks of the Mojave and Arizona terranes, and are common in the terrestrial and shallow water clastic and carbonate rocks of the Apache Group in Arizona and the Pahrump Group of southeastern California (Shride 1967; Hammond and Wooden 1990; Wrucke 1989; Howard 1991). Typically, the diabases form sills where they have intruded sedimentary strata; dikes, although present, are relatively uncommon (Shride 1967; Howard 1991). Studies of diabase from crystalline basement in the relatively undeformed Colorado Plateau province and structural reconstruction of tilted crustal blocks in the Basin–Range confirm that most bodies were originally emplaced as horizontal to subhorizontal sheets (Howard 1991).

The mineralogy and petrology of the diabase intrusions have been described by numerous authors including Shride (1967), Nehru and Prinz (1970), Banks and Krieger (1977), Smith (1970), Hammond (1983, 1986), and Wrucke (1989). Typi-

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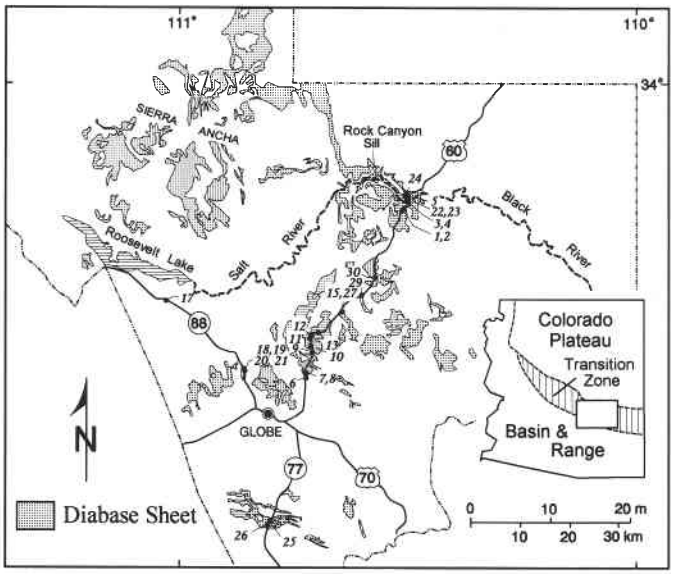


FIG. 1. Geologic sketch map showing the location of diabase sheets in southern Gila County, central Arizona, and the individual paleomagnetic sites sampled in this study. Inset shows the location of the Colorado Plateau, transition zone, and Basin and Range structural and physiographic provinces. Map modified from Wilson et al. (1959).

cally, the diabase is dark gray to green, holocrystalline, medium- to coarse-grained, and displays ophitic to subophitic textures. Plagioclase and clinopyroxene (augite) are the dominant primary phases, but some sills contain up to 5–10% olivine or quartz and potassium-feldspar in graphic intergrowths. Granophyric zones have been reported from some sills (Shride 1967). Accessory phases include minor biotite and hornblende and the opaque phases magnetite, ilmenite, and sulfides (pyrite, chalcopyrite, and pyrrhotite). The diabase sheets are commonly partially to completely altered and replaced by secondary phases. Olivine, where originally present, is altered to serpentine and magnetite, or talc, iddingsite, or bowlingite. Pyroxene is altered to blue-green amphibole and chlorite, or uralite, whereas plagioclase is replaced by clay minerals or carbonate, epidote, and clinozoisite (Shride 1967; Smith 1970; Banks and Krieger 1977). The alteration is thought to be largely deuteric in origin and not the result of subsequent metamorphic, hydrothermal, or tectonothermal events (Shride 1967; Smith 1970; Wrucke 1989; Hammond and Wooden 1990).

Geochemical studies by Hammond (1983), Hammond and Wooden (1990), and Fouts (1974) have characterized the diabase composition as transitional between alkali basalt and tholeiitic basalt. The diabase exhibits low SiO_2, elevated K, Na, Ti, and P values, and moderate light rare earth element (REE) enrichment. An overall iron-enrichment trend is apparent (Hammond and Wooden 1990). Ba/Zr versus P/Zr discrimination diagrams identify two distinct geochemical groups (Hammond 1989), but overall geochemical similarity, geologic setting, and available geochronologic data (summarized in Table 1) from throughout the province (Hammond and Wooden 1990; Shastri et al. 1991; Heaman and Grotzinger 1992) indicate that the intrusions are correlative and were emplaced at about 1070–1100 Ma. Although we consider 1090 Ma to best represent the age of the Arizona diabase, none of the sills studied in this paleomagnetic investigation have been dated by isotopic methods.

The tectonic regime under which the diabase was emplaced remains problematic. Dickinson (1989) has argued that the substantial thickness of the Middle Proterozoic Apache Group sedimentary rocks, the presence of coarse clastics, and intercalated basalt flows implies deposition in an extensional tectonic regime. Stewart (1976), Wright et al. (1976), and Davis et al. (1981) have suggested that deposition of the Apache Group sediments occurred within a rift or aulacogen setting. In contrast, Howard (1991) has argued that widespread diabase intrusion as horizontal sheets required a compressive, or at least isotropic, state of stress. Hammond and Wooden (1990) have proposed that the areally extensive sills of the southwest U.S.A. diabase province should be regarded as the intrusive equivalent of flood basalts or other regionally extensive Proterozoic dike complexes.

Previous paleomagnetic investigations

Paleomagnetic studies of Middle Proterozoic diabase sheets of the southwest U.S.A. diabase province have been conducted by several authors including Helsley (1965), Helsley and Spall (1972), Spall and Troxel (1974), and Sandberg and Butler (1986). Only the results of the two most detailed investigations are described below.

Helsley and Spall (1972) conducted a paleomagnetic investigation of diabase sheets exposed in the Globe – Salt River Canyon region of central Arizona to facilitate the comparison of southwestern U.S. paleomagnetic data with the numerous poles from the Lake Superior region. Eight sites (3–24 samples per site) were collected from sheets demonstrated to be horizontal, thus obviating the need for structural corrections. Progressive alternating field (AF) or thermal demagnetization was carried out in four to five steps at peak fields of up to 140 mT and to maximum temperatures of 600°C; site mean directions were calculated from the demagnetization step that minimized within-site dispersion based on the best estimate of the precision parameter (i.e., Fisher's k; Fisher (1953)). Sites were found to be well-grouped following either AF or thermal demagnetization; α_{95} values ranged from 2 to 6° and k values exceeded 90 for each site. Based on demagnetization results, Curie temperature determinations of 580–590°C, and petrographic examination, Helsley and Spall (1972) concluded that the bulk of the diabase remanence was carried by low-Ti titanomagnetite that probably formed during deuteric alteration. A grand mean direction based on eight sites is $D = 285°$, $I = 49°$ ($k = 56$, $\alpha_{95} = 7°$); the corresponding pole (27°N, 179°E, $\delta p = 6°$, $\delta m = 9°$) is in good agreement with poles from the Lake Superior region, and Helsley and Spall (1972) concluded that there has been no relative displacement between Arizona and the Canadian Shield since about 1150 Ma.

Sandberg and Butler (1986) conducted an investigation of the paleomagnetism of a diabase sill exposed in the Rock Canyon monocline in Salt River Canyon to determine whether monoclines in the Arizona transition zone were Precambrian structures or the result of Laramide (Late Cretaceous – early Tertiary) contraction. They collected 10 sites (six to eight samples per site) from both limbs of the monocline and performed a paleomagnetic fold test of directions from the sill. Pilot samples from each site were progressively demagnetized in a series of steps to peak fields of 80 mT. The remaining samples were treated with a single AF demagnetization step based on the AF field necessary to isolate the characteristic remanent magnetization of the pilot specimens. Following tilt corrections, the site mean directions were characterized by

TABLE 1. Summary of isotopic age determinations reported from tabular intrusions and related rocks of the southwestern U.S.A. diabase province, southeastern California and Arizona

	Location	Method	Material	Age (Ma)	Reference
1	Sierra Ancha	U–Pb	Zr(Gr)	1075±50	Silver (1960)
2	Sierra Ancha	U–Pb	Zr(Gr)	1100±30	Silver (1960, 1963)
				1120±10	Silver (1978)
3	Sierra Ancha	U–Pb	Zr, Bd	1100±2	Shastri et al. (1991)
4	Sierra Ancha	K–Ar	Biotite	1140±30	Damon et al. (1962); Wrucke (1989)
5	Miami, AZ	K–Ar	Biotite	1140±40	Banks et al. (1972)
6	Little Dragoon Mountains	Pb–Pb	Zr(Peg)	1100±15	Silver (1978)[a]
7	Death Valley	U–Pb	Bd	1087±3	Heaman and Grotzinger (1992)
8	Death Valley	U–Pb	Bd	1069±3	Heaman and Grotzinger (1992)
9	Hualapai Mountains	K–Ar	WR	1070±25	H. Stensrud, personal communication in Hammond and Wooden (1990)
10	Grand Canyon	Rb–Sr	WR	1153±30	Hendricks (1972)
11	Grand Canyon	Rb–Sr	WR	1070±30	Hendricks (1972)
12	Grand Canyon	Rb–Sr	WR	1090±90	McKee and Noble (1976)
	(Cardenas Lavas)			1070±70	Elston and McKee (1982)

NOTE: Zr, zircon; Gr, granophyre; Bd, baddelyite; Peg, pegmatite; WR, whole rock.
[a]Silver (1978) suggests that this date is probably best considered a minimum age for emplacement.

moderate to steep negative inclinations. A fold test based on all sites was positive at the 90% confidence level, but two groups of reverse polarity directions were obtained (from the margin and center of the sill) that were significantly different at the 95% confidence level. Sandberg and Butler (1986) suggested that both groups are thermoremanent magnetizations (TRM's) associated with cooling of the sill, but that the center and margins of the sill preserve two different records of secular variation. The presence of a concordant, 2 m thick plate of Middle Proterozoic Dripping Spring Quartzite, isolated within the Rock Canyon sill, was noted by Sandberg and Butler (1986). Such tabular plates of sedimentary strata are relatively common in some Arizona diabase sheets, and in most cases, it can be demonstrated that isolation of such plates has been caused by multiple injections of diabase (Shride 1967). Evidence for a composite intrusion was not described by Sandberg and Butler (1986), but may provide an alternative explanation for the two distinct magnetizations observed in the Rock Canyon sill; the two directional groups may represent spot readings of the geomagnetic field recorded by the intrusion of two separate sills during a single reverse polarity chron. A fold test based on the six interior sites was significant at the 95% confidence level using the statistical test of McFadden and Jones (1981) and indicates that sill intrusion clearly predated Laramide folding. The two group mean directions (D = 271.5°, I = −80.1° and D = 32.5°, I = −77.7°; recalculated using data from Sandberg and Butler (1986) (Table 2, this study)) yield virtual geomagnetic poles (VGP's) at 31.2°N, 272.0°E and 13.3°N, 239.6°E. Sandberg and Butler (1986), noting that Helsley and Spall (1972) observed only normal polarity directions, concluded that the Rock Canyon monocline contained a primary TRM and that diabase intrusion in central Arizona occurred during at least two intervals of opposite polarity.

Paleomagnetic sampling and methodology

Six to fifteen samples were collected at 27 sites in diabase sheets exposed in Gila County, central Arizona (Fig. 1). All sites were collected from roadcut exposures where reasonably fresh cores could easily be obtained. Six sites were collected from sills in horizontal sedimentary rocks exposed in the walls of Salt River Canyon; fourteen sites were collected along U.S. 60 between Salt River Canyon and Globe; five along Arizona State Highway 88 northwest of Globe; and two sites were collected in the Mescal Mountains southwest of Globe (Fig. 1). With the exception of those sites in the Mescal Mountains, all sites are located within or adjacent to the relatively undeformed Colorado Plateau and transition zone and have not been corrected for tilt; the Mescal Mountain sites were corrected for a 30° westward tilt about a horizontal axis of 250°. This structural correction, based on the mean attitude of sedimentary strata adjacent to the sampling sites, is based on geologic mapping by Cornwall and Krieger (1978). Because recent structural and paleomagnetic studies have demonstrated a complex rotational history for the Colorado Plateau region with respect to the stable North American craton (Hamilton 1989; Steiner 1986, 1988; Bryan and Gordon 1986, 1990), the summary paleomagnetic poles from this study have been corrected for a net 5° vertical axis rotation by subtracting 5° from the site mean declinations of sites located within the Colorado Plateau and transition zone.

In the laboratory, each sample was cut into one or more 2.2 cm long specimens. The natural remanent magnetization (NRM) was measured using either spinner or cryogenic magnetometers. Each sample was subjected to detailed progressive AF or thermal demagnetization. Demagnetization results were analyzed using a combination of orthogonal vector diagrams, normalized intensity plots, and equal area projections. Characteristic directions (defined by three or more colinear points that trend toward the origin on a vector diagram) and planar segments (representing the overlapping of two or more components of similar coercivity and (or) unblocking temperature spectra) were determined using principal component analysis (PCA) (Kirschvink 1980). Site mean directions were calculated following Fisher (1953). For site GD21, from which both planar and linear data were obtained, the mean direction was calculated using method two of Kirschvink (1980) and Bingham statistics (Onstott 1980) were calculated. Magnetic carriers were investigated by acquisition of isothermal remanent magnetization (IRM) experiments and Curie temperature determinations.

TABLE 2. Paleomagnetic site mean data from diabase sills of the southwestern U.S.A. diabase province, Gila County, central Arizona

No.	Site	*n*/*N*	*D* (°)	*I* (°)	*R*	*k*	α_{95} (°)	Slong. (°)	Slat. (°)	Plong. (°)	Plat. (°)
						Normal polarity sites					
	GD01	9/9	281.3	58.2	8.97	268.2	3.2	249.5	33.7	189.3	28.4
	GD02	8/8	277.3	55.7	7.96	165.8	4.3	249.5	33.7	188.0	24.4
1	Avg.		279.2	57.0						188.6	26.4
2	GD03	11/11	285.3	51.4	10.94	165.0	3.6	249.5	33.8	180.7	28.8
3	GD04	7/8	295.5	56.6	6.93	81.9	7.7	249.5	33.8	182.9	38.4
4	GD05	7/7	283.2	26.0	6.94	108.5	5.8	249.5	33.8	163.9	18.4
	GD06	9/10	301.8	54.9	8.79	38.6	8.4	249.3	33.5	178.9	42.7
	GD07	9/9	268.6	40.6	8.97	265.4	3.2	249.3	33.5	179.7	11.4
5	Avg.		282.8	48.9						179.4	27.1
6	GD08	13/14	318.0	61.2	12.79	56.1	5.6	249.3	33.5	186.8	56.1
7	GD09	9/10	299.2	53.8	8.62	21.1	11.5	249.3	33.5	178.3	40.3
8	GD10	6/6	267.5	38.2	5.99	421.5	3.3	249.3	33.5	178.7	9.7
	GD12	4/7	283.7	44.3	3.97	94.2	9.5	249.3	33.5	175.2	24.8
	GD13	10/10	281.4	42.7	9.98	427.0	2.3	249.3	33.5	175.0	22.4
9	Avg.		282.5	43.5						175.1	23.6
10	GD17	10/12	285.9	17.1	9.74	34.5	8.3	249.1	33.6	157.8	18.0
	GD18	8/9	345.6	35.2	7.98	386.0	2.8	249.2	33.5	115.6	71.0
	GD19	5/5	343.7	34.1	4.97	119.6	7.0	249.2	33.5	117.8	69.3
	GD20	8/8	328.8	38.9	7.97	242.4	3.6	249.2	33.5	144.5	60.2
11	Avg.		339.6	36.3						128.4	67.4
12	GD22	8/9	270.1	40.4	7.94	120.3	5.1	249.5	33.8	178.9	12.7
13	GD23	9/9	277.6	0.0	8.95	153.8	4.2	249.5	33.8	155.3	6.3
14	GD24	9/10	297.9	58.5	8.95	168.0	4.0	249.5	33.8	184.8	40.8
15	GD25[a]	9/11	272.9	52.0	8.62	21.0	11.5	249.2	33.2	186.2	19.3
16	GD26[a]	6/9	285.2	45.6	5.95	99.6	6.8	249.2	33.2	175.6	26.4
17	GD29	9/9	306.9	66.4	8.99	678.5	2.0	249.4	33.6	197.2	48.2
18	GD30	6/6	264.1	33.4	5.97	157.6	5.4	249.4	33.6	177.8	5.3
						Reverse polarity sites					
19	GD11[b]	8/8	115.3	−69.9	7.96	173.7	4.2	249.3	33.5	204.3	41.0
	GD15	8/9	232.7	−71.9	7.92	82.8	6.1	249.4	33.6	289.7	47.7
	GD27	8/10	215.5	−75.1	7.99	484.5	2.5	249.4	33.6	276.9	53.8
20	Avg.		224.9	−73.7						283.8	50.9
21	GD21[c]	8/12	195.7	−52.4		−60.8/−0.7[d]	5.9/7.1[e]	249.2	33.5	336.4	75.4
22	SR #1[f]	4/4[g]	271.5	−80.1	3.99	200.5	5.6	249.4	33.6	272.0	31.2
23	SR #2[f]	3/3[g]	32.5	−77.7	2.99	154.2	7.1	249.4	33.6	239.6	13.3

NOTES: No., independent observation number, e.g., "cooling unit"; Avg., mean of two or more sites used to establish an independent direction; *n*/*N*, number of samples used in the calculation of the site mean direction/samples collected and measured; *D*, mean declination; *I*, mean inclination; *R*, length of the resultant vector; *k*, precision parameter (Fisher 1953); α_{95}, semiangle of the cone of 95% confidence about the site mean direction; Slong. and Slat., longitude and latitude of the site locality, respectively; Plong. and Plat., longitude and latitude of the virtual geomagnetic pole, respectively. Note that VGP's for the reverse polarity sites have been calculated following inversion to normal polarity.

[a]Site mean direction after correction for a tilt of 250°/30°W; site mean data from GD25 are based on high temperature component uncovered during thermal demagnetization.

[b]Site mean direction calculated from the high coercivity remanence only, as discussed in the text.

[c]This site is suspected of having been remagnetized by a nearby Tertiary pluton and the observed direction is not related to the reverse polarity direction characteristic of some diabase sheets.

[d]Bingham concentration paramaters k_1 and k_2 (Onstott 1980).

[e]Semiangles of the minor and major axes of the Bingham ellipse of 95% confidence about the site mean direction (Onstott 1980).

[f]Mean directions calculated from the two statistically distinct site mean populations reported in Sandberg and Butler (1986).

[g]Number of sites used in the calculation of mean direction for the data of Sandberg and Butler (1986).

Results

Demagnetization of diabase samples by either AF or thermal methods was relatively straightforward. NRM intensities ranged from 0.00872 to 22.4 $A \cdot m^{-1}$ and averaged 1.6 $A \cdot m^{-1}$. Most samples contained one or two magnetic components. The first component removed is usually a minor secondary magnetization of northerly declination and moderate positive inclination and is aligned approximately subparallel with the Earth's present field direction. This magnetization is commonly removed by AF demagnetization of 5−10 mT and by thermal demagnetization in temperatures less than 300°C. The direction and magnetic characteristics (i.e., low coercivity and unblocking temperatures) of this component are consistent with an origin as a viscous remanent magnetization acquired in the present day field.

Following removal of the secondary magnetization where present, the majority of diabase samples exhibited behavior characterized by the linear decay of the magnetic vector to the origin of a demagnetization diagram. Such behavior is usually indicative of the removal of a single magnetization component, and we term this component the characteristic magnetization (ChRM) of the sample. Two distinct ChRM's were observed: those of west to northwest, moderate positive incli-

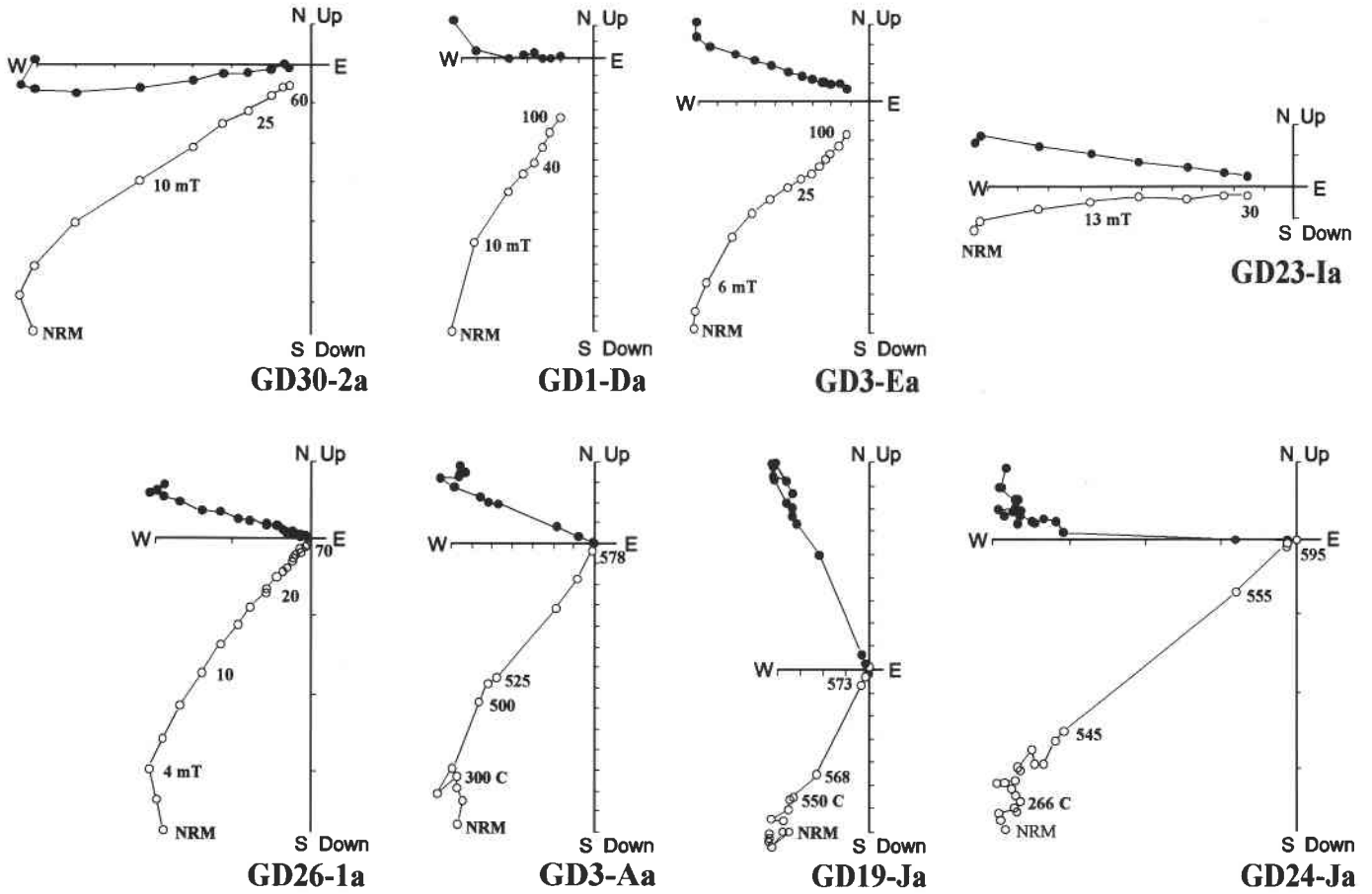


FIG. 2. Orthogonal vector diagrams showing representative examples of demagnetization behavior of diabase samples that carry the ADn magnetization. Open and closed symbols represent the magnetization components projected onto the vertical and horizontal planes, respectively. Demagnetization in millitesla and degrees Celsius. Plot scale (all diagrams): 1 division = 0.1 $A \cdot m^{-1}$.

nation (termed ADn); and those of steep negative inclination (termed ADr).[1] The ChRM's for both groups of samples were isolated in demagnetization between 10 and 90 mT or temperatures of 300−580°C. Negative inclination samples commonly displayed slightly greater coercivities than normal polarity samples, but both groups are characterized by a relatively narrow range of laboratory unblocking temperatures. Representative demagnetization diagrams illustrating AF and thermal demagnetization of both groups are shown in Figs. 2 and 3. ChRM directions for individual specimens were well-defined using PCA with anchored maximum angular deviation (MAD) values ranging from 1.0 to 10.0°; samples with MAD values >10.0° were not used in the calculation of site mean directions. Site mean directions are well-defined, with Fisher's k values ranging from 21 to 679 and values of α_{95} from 2.0 to 11.5°, with most about 3 to 5° (Fig. 4; Table 2). Where two or more sites were collected from the same sill and (or) stratigraphic level, and were expected to provide essentially identical recordings of the geomagnetic field, their site mean directions have been averaged to provide "cooling unit" mean directions (e.g., sites GD15 and GD27 in Table 2). This method, however, may be expected to slightly underestimate the precision by which the cooling unit mean is known, owing to the small number of sites used.

[1]ADn and ADr refer to the "Arizona diabase"; the designations "n" and "r" indicate normal and reverse polarity magnetizations in present-day coordinates.

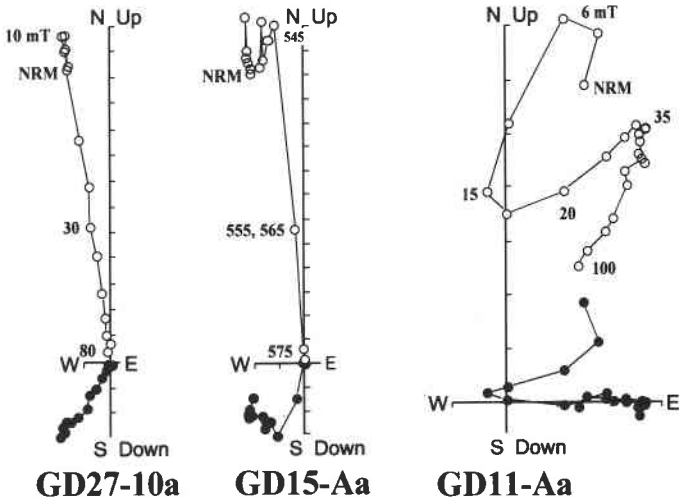


FIG. 3. Orthogonal vector diagrams showing representative examples of the demagnetization behavior of diabase samples that carry the ADr magnetization. Conventions as in Figure 2. Plot scale (all diagrams): 1 division = 0.1 $A \cdot m^{-1}$.

At three sites, we observed more complicated or unusual results. Samples from site GD11 contain at least three distinct magnetizations that are readily isolated in AF demagnetization (Fig. 3), but cannot be as readily resolved by thermal demag-

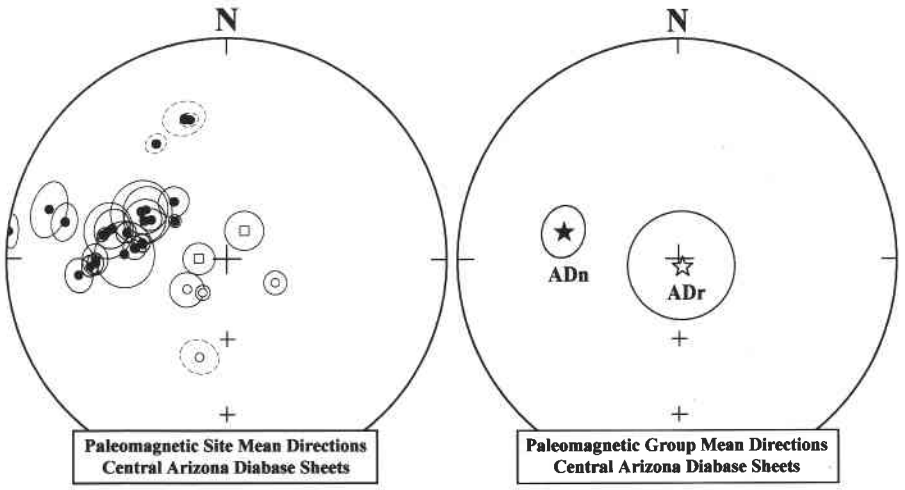


FIG. 4. Equal area projections showing the paleomagnetic site mean directions and their α_{95} cones of confidence. Site mean directions with dashed α_{95} cones are sites excluded from the calculation of the group mean directions. Circles are site means determined from this study; squares are the means of the two independent directions reported by Sandberg and Butler (1986). Solid circles are normal polarity directions; open circles are reversed polarity directions. ADn and ADr are the observed normal and reverse polarity group mean directions.

netization. A low-coercivity component, revealed by AF demagnetization in peak fields between 3 and 20 mT, is characterized by a steep negative inclination direction. AF demagnetization between fields of 15 to 40 mT isolates a component of northwesterly declination and moderate positive inclination, similar to the ADn ChRM derived from the majority of the diabase sites. At fields greater than 35–40 mT, the sample demagnetization trajectories are characterized by linear decay toward the origin of their respective demagnetization diagrams, and a magnetization of steep negative inclination is well-resolved. At present we have no explanation for this complicated behavior, but to speculate that this multicomponent remanence may reflect a complex cooling history, possibly the result of intrusion of nearby sills of different polarity. For this report, only the high-coercivity remanence from site GD11 has been used to calculate a site mean direction.

At site GD25, in the Mescal Mountains southwest of Globe (Fig. 1), most samples collected were heavily iron stained and contain a magnetization of high coercivity (up to 80 mT) but relatively low (i.e., less than 350–400°C) laboratory unblocking temperatures. The in situ (uncorrected for tilt) site mean direction for this remanence is $D = 179.1°$, $I = -54.4°$ ($k = 40.2$, $\alpha_{95} = 8.2°$, $n = 9$ samples). This direction, although similar to the steep negative inclination direction observed in other sites, is closer to the expected direction for this region for the Late Cretaceous or Tertiary. After demagnetization to temperatures of 400–580°C, a magnetization of northwesterly declination and moderate positive inclination is isolated. After tilt correction, this ChRM is similar to those obtained from other diabase sites (i.e., $D = 272.9°$, $I = 52.0°$; Table 2). The secondary magnetization at this site may be the result of alteration caused by hydrothermal fluids associated with the widespread Cretaceous and Tertiary intrusions present in the Mescal Range; if correct, then acquisition of this secondary magnetization may have largely postdated either Laramide or Basin and Range tilting of the Mescal Range.

Finally, at site GD21 northwest of Globe (Fig. 1), a distinct ChRM could not be isolated by either AF or thermal demagnetization. Rather, two components of overlapping coercivity and laboratory unblocking temperature spectra were present, resulting in planar demagnetization trajectories that converged toward a negative inclination direction (Fig. 4; Table 2). Because this site is from a structurally complex area and we suspect it of being affected by Late Cretaceous or Tertiary remagnetization, this site has not been used in the calculation of a group mean direction.

The demagnetization characteristics of the diabase sheets, with the exception of the secondary magnetization at site GD25 described above, indicate that the primary carrier in the diabase sills is a low-coercivity phase, such as magnetite. Rapid acquisition of IRM and saturation in fields less than 200 mT confirm this (Fig. 5). Strong field thermomagnetic experiments from this study yielded heating and cooling curves that are typically reversible, or nearly so, and gave Curie temperatures that ranged from 530 to 570°C ($n = 6$ samples) (Fig. 6), slightly lower than the temperatures of 570–580°C reported by Helsley and Spall (1972) and Sandberg and Butler (1986). The results from all studies indicate that low-Ti titanomagnetite is the dominant magnetic phase in both the normal and reverse polarity diabase. The wide range of coercivities, and similarity between normalized decay curves for IRM, AF, and direct current (dc) demagnetization of IRM (Fig. 5) suggest that the remanence is carried by a mixture of multidomain and psuedosingle- or single-domain magnetite. With the exception of slightly greater coercivities noted previously, there are no significant petrologic or rock magnetic differences between sites that contain normal and reverse polarity magnetizations.

Discussion

Paleomagnetism

Paleomagnetic results from this study of the central Arizona diabase sheets yield two distinct remanent magnetizations, in accordance with the findings of previous investigations (Helsley and Spall 1972; Sandberg and Butler 1986). The first magnetization (ADn), defined by 18 independent observations, is relatively well-defined and is of northwest declination and moderate positive inclination (i.e., $D = 286.6°$, $I = 45.4°$, $k = 14.7$, $\alpha_{95} = 9.3°$) (Table 3). After eliminating cooling unit 11 (Table 2), which seems to record an unusual northerly

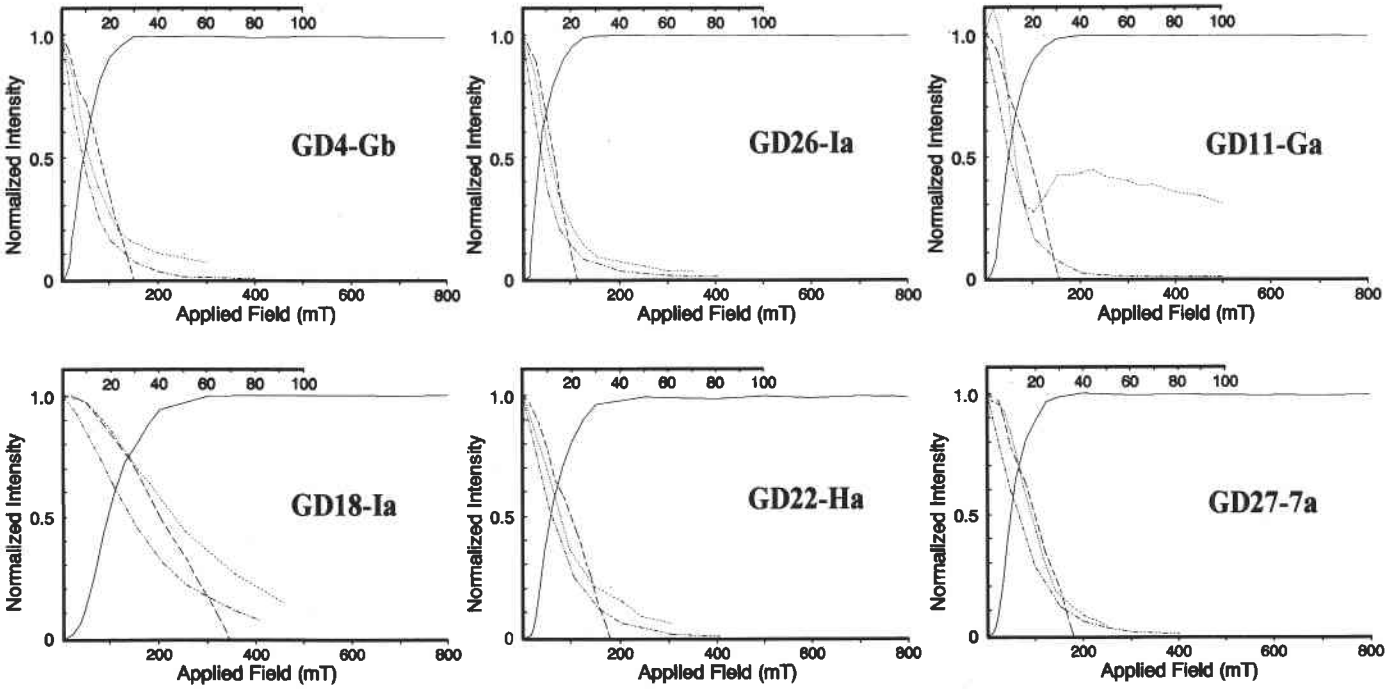


FIG. 5. Normalized intensity diagrams of diabase samples that show the acquisition of isothermal remanent magnetization (IRM) (solid line; lower scale) induced by the application of an applied dc field; alternating field (AF) demagnetization of natural remanent magnetization (NRM) (dotted line; upper scale); AF demagnetization of IRM (dotted−dashed line; upper scale); and dc demagnetization of saturation IRM (dashed line; upper scale). Samples GD27-7a and GD11-Ga are samples that carry the ADr magnetization, all others are ADn.

declination and is located within a structurally complex area associated with Late Cretaceous − Tertiary magmatism, we obtain a preferred mean direction of $D = 283.3°$, $I = 45.1°$ ($k = 17.7$, $\alpha_{95} = 8.7°$) (Table 3; Fig. 4). Both mean directions are indistinguishable at the 95% confidence level from the previously reported group mean direction of Helsley and Spall (1972). Correction of site mean directions for a net 5° clockwise rotation of the Colorado Plateau and the transition zone (Bryan and Gordon 1990) gives $D = 278.9°$, $I = 45.1°$ ($k = 17.9$, $\alpha_{95} = 8.7°$) (Table 3). A mean paleomagnetic pole, calculated assuming an axially geocentric dipole and mapping site mean directions into VGP space, is located at 22.7°N, 179.3°E ($K = 21.9$, $A_{95} = 7.8°$) (Table 3) in the north central Pacific Ocean (Fig. 7).

The observed scatter in the ADn paleomagnetic site mean directions, based on the best estimate of the precision parameter k of the group mean direction (Table 3), is significantly greater than that reported by Helsley and Spall (1972) for their study of the central Arizona diabase (i.e., 17.7 vs. 57). This suggests that a greater amount of directional scatter is present within the data obtained in this study. Such dispersion may have arisen by several different means. First, some scatter in site mean directions may have resulted from failure to account for unrecognized postemplacement tilt of the diabase sheets sampled in this study. In the investigation of Helsley and Spall (1972), samples were collected only where it could be demonstrated that the diabase sheets were horizontal; sampling for this study was not as restrictive and therefore the potential exists that some of the sheets sampled may have experienced slight, probably $< 10°$ tilting. Diabase sheets exposed in the walls of Salt River Canyon, however, where Proterozoic sedimentary rocks are essentially flat lying, exhibit a large range of declinations and inclinations from individual sites (i.e.,

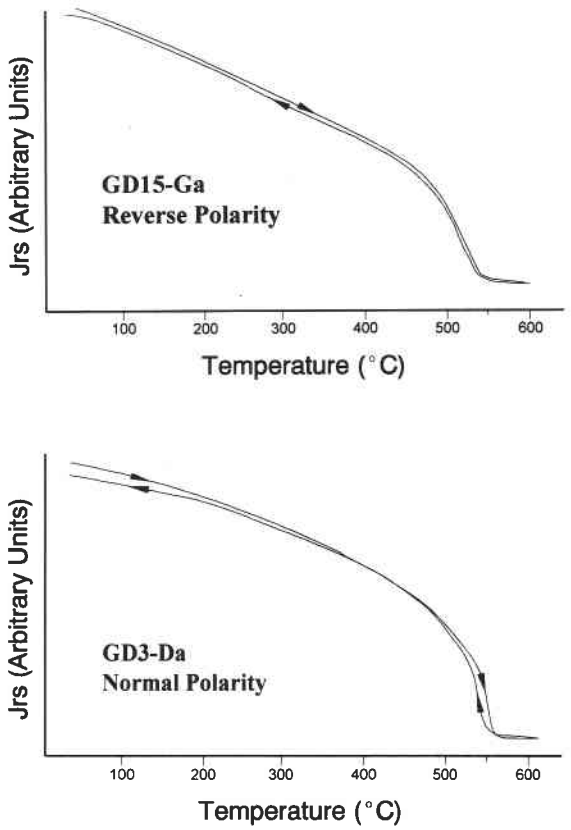


FIG. 6. Representative strong-field thermomagnetic determinations of Curie temperatures from diabase samples that show both normal and reverse polarity magnetizations. Samples run in air with a saturating field of 0.2 T.

TABLE 3. Paleomagnetic pole positions calculated from site mean data from diabase sheets of the southwestern U.S.A. diabase province, Gila County, Arizona

Direction statistics									Pole statistics				
D (°)	I (°)	N	k	α_{95} (°)	Plong. (°)	Plat. (°)	δp (°)	δm (°)	Plong. (°)	Plat. (°)	K	A_{95} (°)	δ_{63} (°)
Normal polarity, ADn													
286.6[a]	45.4	18	14.7	9.3	174.8	27.5	7.5	11.8	176.2	29.0	15.8	9.0	20.4
283.3[b]	45.1	17	17.7	8.7	176.0	24.8	7.0	11.0	177.3	26.5	21.9	7.8	17.4
Normal polarity, ADn (corrected for proposed 5° clockwise rotation of the Colorado Plateau)[c]													
282.2[a]	45.3	18	14.7	9.3	176.6	24.0	7.5	11.8	178.0	25.2	15.7	9.0	20.5
278.9[b]	45.1	17	17.9	8.7	178.0	21.3	7.0	11.0	179.3	22.7	21.9	7.8	17.3
Reverse polarity, ADr													
161.1	−87.5	4	22.2	19.9	247.3[d]	38.6	39.7	39.8	248.1	38.0	6.6	38.9	32.0
Reverse polarity, ADr (corrected for proposed 5° clockwise rotation of the Colorado Plateau)[c]													
155.2	−87.5	4	22.1	20.0	246.7[d]	38.1	39.9	39.9	247.6	37.6	6.5	38.9	32.1

NOTES: D, declination; I, inclination; N, number of independent directions used in the calculation of group mean directions and paleomagnetic poles; k, precision parameter; α_{95}, semiangle of 95% confidence of the group mean direction; Plong. and Plat., longitude and latitude of the mean paleomagnetic pole, respectively (the group mean pole has been calucated by converting the mean direction into pole space, the paleomagnetic pole has been calculated by converting cooling mean directions into VGP space and calculating a mean pole using Fisher (1953) statistics); δp and δm, minor and major semiaxes of the 95% error ellipse about the pole position, respectively; K, precision parameter of the virtual geomagnetic pole; A_{95}, semiangle of 95% confidence of the paleomagnetic pole; δ_{63}, angular standard deviation.

[a]Group mean direction calculated using all cooling units.

[b]Group mean direction calculated after exclusion of cooling unit 11 that we suspect may have been remagnetized by Late Cretaceous − Tertiary magmatism.

[c]All cooling unit directions, except 15 and 16 of Table 2, have been corrected subtracting 5° from the site mean and cooling unit mean directions in Table 2 to account for the current best estimate of Bryan and Gordon (1990) for clockwise rotation of the Colorado Plateau; cooling units 15 and 16 are clearly outside of the Colorado Plateau and the transition zone and were not corrected for rotation.

[d]Pole position calculated after inversion of the reverse polarity data to normal polarity.

declinations range from 270 to 302° and inclinations from 0 to 59°; Table 2). Such dispersion clearly cannot have resulted from tilting. Elsewhere, sites where structural control is lacking generally give directions that are comparable with the range observed from sites within the canyon, suggesting that the effects of unrecognized tilting are probably minor.

Another possible source of scatter may be from failure to fully isolate the ChRM of the diabase. This appears unlikely given the ready elimination of obvious low-coercivity overprints in either AF or thermal demagnetization and the generally well-defined ChRM exhibited by most samples. The most probable source of a contaminating overprint is that of viscous remanent magnetization (VRM) induced by the present-day field. Failure to remove such a VRM would probably result in a systematic bias of individual sample directions toward steeper inclinations and more northerly declinations, but most ChRM and site mean directions are significantly removed from that of the present-day field. In cases where contamination of the NRM by Laramide or younger igneous activity appears to have been a significant problem (i.e., sites GD19-22 northwest of Globe, Fig. 1), we have eliminated those data from the group mean calculation. Failure to fully isolate the diabase ChRM does not appear to be a significant source of dispersion in this study.

Given that postemplacement tilting or failure to isolate the ChRM are unlikely causes of significant dispersion in the ADn directions, and assuming that the ADn magnetization represents a TRM acquired during emplacement and cooling of the sheets, the most probable cause of scatter of the site mean directions may be attributed to secular variation of the geomagnetic field. Calculation of a mean paleomagnetic pole from the 17 independent VGP's (described above) gives an angular standard deviation (δ_{63}) of 17.3° (Table 3) with upper and lower confidence limits of 5.2 and 3.3°, respectively (Cox 1969). The expected VGP scatter (S_λ) for 26.7° (the calculated paleolatitude for central Arizona based on the observed paleomagnetic data) using model G of McFadden et al. (1991) for the interval 0−5 Ma is 14.2 ± 0.75°. The observed value for the diabase is slightly greater than the predicted value, but the two are indistinguishable at the 95% confidence level. The observed value is similar, although slightly greater, than results from the Keweenawan region (Halls and Pesonen 1982) and compares favorably with predicted values for most of the middle and late Cenozoic (i.e., 18.5° for 5−22.5 Ma, 17.2° for 22.5−45 Ma), but is somewhat greater than that for the early Cenozoic and Late Cretaceous (13.3° for 45−80 Ma, 11.0° for 80−110 Ma). This result is interpreted to indicate that most of the scatter in the Arizona diabase sheets is the result of recording the normal range of secular variation of the geomagnetic field during emplacement of the diabase sheets. Thus, intrusion probably occurred over a period greater than 10^5 or 10^6 years to adequately sample this range of secular variation. It must be noted that expected ranges of secular variation for the Precambrian are unknown and models of secular variation based on recent geomagnetic field behavior may not be strictly applicable to the analysis of Precambrian paleomagnetic data. Halls and Pesonen (1982), however, have suggested that the observed range of secular variation for the extensively studied Keweenawan rocks compares favorably with that of the present day suggesting that such an approach may have some validity. Finally, although the observed dispersion in the diabase sheets is consistent with that predicted from recent analyses of secular variation, and we conclude that paleosecular variation has been adequately averaged, contributions to the overall dispersion from other causes such as those discussed above cannot be entirely discounted.

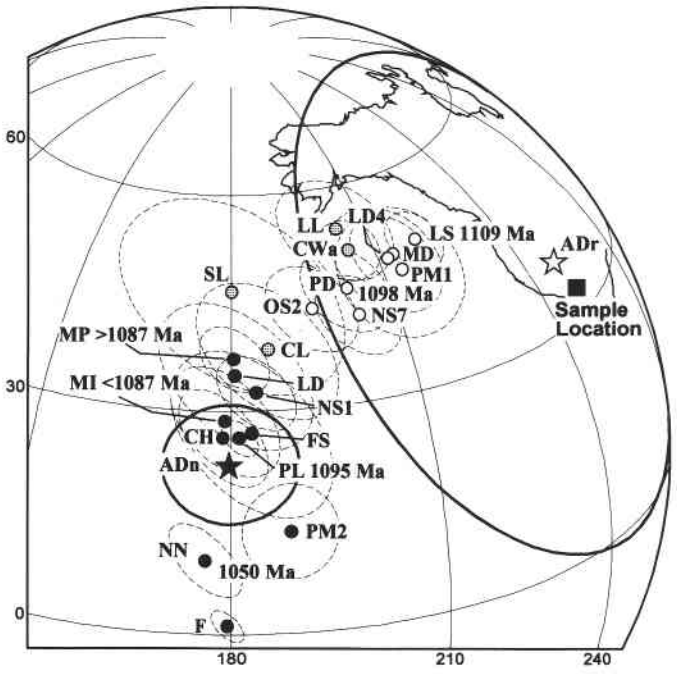


FIG. 7. Orthographic global projection showing paleomagnetic poles that define the ca. 1100 Ma western arm of the Logan Loop of the North American apparent polar wander (APW) path and the location of poles from normal and reverse polarity diabase from central Arizona. Poles and abbreviations for most poles are selected from the compilation of Halls and Pesonen (1982): F, Freda Sandstone; LD4, Logan dikes (reversed); LS, Logan sills (mean pole); MD, Marquette dikes; MI, Michipicoten Island Volcanics; NN, Nonesuch Shale; NS7, Lower North Shore Volcanics; OS2, upper part of the Osler Sequence (reversed); PL, Portage Lake Volcanic Group (mean pole). Poles PM1 and PM2 are revised poles from the Powder Mill Group (Palmer and Halls 1986). Pole FS is from the Firesand River Carbonatite Complex (Symons 1989*a*); CL and SL are poles from the Chipman Lake and Seabrook Lake Carbonatite complexes, respectively (Symons 1992); LL, Lackner Lake complex (Symons 1989*b*); and CH, Clay-Howells Carbonatite Complex (Lewchuk and Symons 1989). Pole CWa is from the Coldwell Complex and is derived from paleomagnetic data reported in Lewchuk and Symons (1990). Because recent U–Pb dating indicates that there are no significant differences in age between the major phases of this pluton (Heaman and Machado 1992), VGP's from all sites (including those of Robertson, unpublished) with α_{95} less than 15° have been combined to yield an average pole at 50.4°N, 204.5°E (K = 32.2, A_{95} = 4.4°). Poles ADn and ADr are normal and reverse polarity poles from this study and have been corrected for a 5° clockwise rotation of the Colorado Plateau; the confidence limits for these poles are denoted by the heavy solid lines, all others are shown as dashed lines. Paleomagnetic poles derived from normal polarity mean directions are filled circles, those derived from reverse polarity data are open, and poles that contain paleomagnetic data of mixed polarity are shaded. Ages assigned to poles are from Davis and Paces (1990), Symons (1989*a*, 1989*b*), and Heaman and Machado (1992).

The relative age of the ADn magnetization with respect to the emplacement age of the diabase sheets cannot be established with certainty because rigorous baked contact tests that would unequivocally demonstrate that the ADn remanence is a primary TRM could not be conducted. The close agreement of the tilted Mescal range sites after structural correction with other sills known or assumed to be flat lying is consistent with a pretilting (i.e., Laramide or Basin and Range) age for remanence acquisition. The absence of igneous rocks of younger age in the vicinity of most of the diabase sites and the discordance between the ADn direction and the present-day field and expected directions for the Late Cretaceous – Cenozoic precludes the possibility of remagnetization. The most compelling evidence for a primary TRM, as noted by previous workers (Helsley 1965; Helsley and Spall 1972), is the consistency of the ADn pole with those of similar age from the paleomagnetically well-defined western arm of the Logan Loop of the midcontinent rift (Fig. 7).

The second magnetization (ADr) is less well defined than the ADn remanence, being represented by only four distinct observations from three sheets (including the Rock Canyon monocline sill of Sandberg and Butler (1986); data recalculated in Table 2 of this study). The ADr magnetization is characterized by an overall southeast in situ mean declination and steep negative inclination (in situ D = 161.1°, I = −87.5°, k = 22.2, α_{95} = 20°) (Fig. 6) with a corresponding VGP, corrected for a 5° clockwise rotation of the Colorado Plateau, at 37.6°N, 247.6°E (K = 6.5, A_{95} = 38.9°) (Fig. 7; Table 3). Because of the small number of observations, a rigorous analysis of the dispersion of site mean data has not been attempted. We note that the ADn and ADr magnetizations are clearly not antipodal, and that the two group mean directions are statistically distinct at the 95% confidence level.

Although the ADn magnetization appears to be a TRM acquired following emplacement and cooling of the diabase at about 1090 Ma, the origin of the ADr remanence is more problematic. The magnetization passes a fold test at 95% confidence level and clearly predates Laramide (Late Cretaceous – early Tertiary) folding (Sandberg and Butler 1986). Beyond this constraint, the age of remanence cannot be established as it has not been possible to conduct baked contact tests to establish if ADr is a primary TRM or has originated by some other means. Sandberg and Butler (1986) concluded that the magnetization obtained from the Rock Canyon monocline sill was a TRM, apparently on the basis of the ready removal of secondary magnetizations during AF demagnetization. They noted that the observed site mean directions from the sill, after structural correction, were significantly different from the results of Helsley and Spall (1972) and suggested that diabase intrusion in central Arizona must have occurred during two intervals of opposite polarity of the geomagnetic field. Similarly, no conclusive evidence regarding the age of magnetization has been determined in this study. At site GD11, however, a low-coercivity positive inclination magnetization similar to the ADn remanence appears to be superimposed upon a high-coercivity magnetization that we consider to be ADr. If it can be assumed that the high-coercivity magnetization is the "primary" magnetization of this site, then the ADn magnetization appears to be younger than ADr; unfortunately, this could not be confirmed by thermal demagnetization. Furthermore, we note that secondary chemical remanent magnetizations or thermo-chemical remanent magnetizations can, under appropriate conditions, generate magnetizations of higher coercivity and higher laboratory unblocking temperatures than the primary remanence (e.g., generation of single domain magnetite from primary titanomagnetite during deuteric or hydrothermal alteration) (Hargraves and Banerjee 1973; Hagstrum and Johnson 1986). In the case of the ADr magnetization, there appears to be no evidence of significantly younger thermal or tectonothermal events in the vicinity of the sites that carry the ADr remanence. As concluded by Sandberg and Butler (1986) in their study of the paleomagnetism of the

Rock Canyon sill, the ADr magnetization observed in this study is probably a TRM acquired during a reverse polarity chron.

Paleomagnetic poles and tectonic implications

Mean paleomagnetic poles for the ADn and ADr magnetizations from this study have been calculated assuming an axially geocentric dipole field and are compared with age-equivalent poles from the northern midcontinent (Fig. 7). Numerous detailed paleomagnetic studies of abundant volcanic, intrusive, and sedimentary rocks of the Keweenawan region of the midcontinent rift have resulted in definition of the western arm of the Logan Loop of the North American apparent polar wander (APW) path. Recent high-precision U–Pb geochronologic data from Keweenawan rocks indicate that magmatism spanned 22 Ma, from 1109 to 1087 Ma, and demonstrated that most of the exposed volcanic section was extruded over a period of about 4 Ma between 1098 and 1094 Ma (Davis and Paces 1990). The transition between predominantly reversed to normal polarity chrons is constrained to have occurred at about 1098 ± 2 to 3 Ma (Paces and Miller 1993), although recent paleomagnetic studies have documented the presence of multiple polarities during both chrons (Symons 1989*b*, 1992; Lewchuk and Symons 1990; Costanzo-Alvarez et al. 1993). The combination of detailed paleomagnetic studies and high-precision geochronologic data has led to a relatively precisely calibrated portion of the Precambrian APW path that rivals or exceeds some portions of the Phanerozoic APW path.

The ADn pole from this study, as noted earlier, is in excellent agreement with several normal polarity poles from the midcontinent rift (Fig. 7), regardless of whether the pole is corrected for clockwise rotation of the Colorado Plateau (Bryan and Gordon 1990). This indicates, as noted previously by Helsley (1965), that there has been no significant displacement between central Arizona and the North American craton since Middle Proterozoic time. Although structural and paleomagnetic evidence has demonstrated slight Laramide-age and Late Cenozoic rotation of the Colorado Plateau, net rotation of the Colorado Plateau since Proterozoic time is relatively minor.

The ADr pole plots in a region of the APW path that is largely devoid of paleomagnetic data, and thus is potentially an important contribution to the paleomagnetic data base. Unfortunately, this pole is poorly constrained due to the small number of sites averaged and by error introduced by mapping of the steep inclination directions into pole space; consequently, the use of the ADr VGP for definition of the APW track is precluded. Identification and paleomagnetic analysis of additional sheets that carry the ADr magnetization may substantially improve the quality of the ADr pole. Reverse polarity sheets in the central Arizona region should be readily distinguishable from normal polarity sheets based on their aeromagnetic signatures.

The close agreement of the ADn pole with those derived from normal polarity magnetizations from Keweenawan rocks of the midcontinent rift is consistent with an age of about 1090–1100 Ma for magnetization acquisition, and this age is compatible with most of the geochronologic data listed in Table 1. The strongly asymmetric nature of the ADr pole, however, may indicate that magmatism in the central Arizona region may have occurred somewhat earlier than the 1090–1100 Ma dates suggested by the available U–Pb isotopic dates. The uniformity between paleomagnetic poles and U–Pb geochronology from the midcontinent rift and the southwestern U.S.A. diabase province indicates that widespread mafic magmatism in both regions was essentially synchronous. Such large-scale, voluminous mafic magmatism within a relatively short period of time probably indicates the initiation and upwelling of one or more mantle plumes in order to cause fragmentation and extension of the continental crust (Heaman et al. 1992; Howard 1991). Additional high-precision U–Pb dating of diabase sheets in the southwestern U.S.A. diabase province, such as that recently reported by Shastri et al. (1991) and Heaman and Grotzinger (1992) is needed, however, to unequivocally demonstrate that magmatism in the two regions are equivalent in age and to demonstrate that all of the diabase sheets are of similar age.

Conclusions

Paleomagnetic results from this study of the central Arizona diabase sheets largely confirm the findings of previous paleomagnetic investigations: two distinct paleomagnetic directions, one of west–northwest declination and moderate positive inclination (ADn), and the other of southeast steep negative inclination (ADr) were obtained. Both magnetizations are interpreted to be TRM's acquired at about 1090 Ma. Close agreement between the well-defined Arizona diabase ADn pole and those of Keweenawan rocks and available geochronologic data indicate that mafic magmatism in the southwestern U.S.A. diabase province and the midcontinent rift was essentially synchronous.

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